Application of Eq. (9.124) specifies the magnitude of the incremental length l:

$$l = \frac{n_0 u_p}{f_0} = \frac{9 \times 0.8 \times 3 \times 10^8}{10^{10}} = 21.6 \text{ cm}.$$

In summary, with $N = 6$ and $kd = \pi$, Eq. (9.121) becomes

$$F_{an}(\gamma') = \frac{\sin^2(3\gamma')}{36 \sin^2(\gamma'/2)}$$

with

$$\gamma' = kd(\cos\theta - \cos\theta_0) = \pi(\cos\theta - \cos\theta_0)$$

and

$$\cos\theta_0 = \frac{2n_0\pi}{kd}\left(\frac{\Delta f}{f_0}\right) = 18\left(\frac{f - 10 \text{ GHz}}{10 \text{ GHz}}\right). \qquad (9.130)$$

The shape of the array pattern is similar to that shown in **Fig. 9-30**, and its main-beam direction is along $\theta = \theta_0$. For $f = f_0 = 10$ GHz, $\theta_0 = 90°$ (broadside direction); for $f = 10.48$ GHz, $\theta_0 = 30°$; and for $f = 9.52$ GHz, $\theta_0 = 150°$. For any other value of θ_0 between 30° and 150°, Eq. (9.130) provides the means for calculating the required value of the oscillator frequency f.

Concept Question 9-11: Why are antenna arrays useful? Give examples of typical applications.

Concept Question 9-12: Explain how the pattern multiplication principle is used to compute the radiation pattern of an antenna array.

Concept Question 9-13: For a linear array, what roles do the array amplitudes and phases play?

Concept Question 9-14: Explain how electronic beam steering is accomplished.

Concept Question 9-15: Why is frequency scanning an attractive technique for steering the beam of an antenna array?

Chapter 9 Summary

Concepts

- An antenna is a transducer between a guided wave propagating on a transmission line and an EM wave propagating in an unbounded medium, or vice versa.
- Except for some solid-state antennas composed of nonlinear semiconductors or ferrite materials, antennas are reciprocal devices; they exhibit the same radiation patterns for transmission as for reception.
- In the far-field region of an antenna, the radiated energy is approximately a plane wave.
- The electric field radiated by current antennas, such as wires, is equal to the sum of the electric fields radiated by all the Hertzian dipoles making up the antenna.
- The radiation resistance R_{rad} of a half-wave dipole is 73 Ω, which can be easily matched to a transmission line.
- The directional properties of an antenna are described by its radiation pattern, directivity, pattern solid angle, and half-power beamwidth.
- The Friis transmission formula relates the power received by an antenna to that transmitted by another antenna at a specified distance away.
- The far-zone electric field radiated by a large aperture (measured in wavelengths) is related to the field distribution across the aperture by Kirchhoff's scalar diffraction theory. A uniform aperture distribution produces a far-field pattern with the narrowest possible beamwidth.
- By controlling the amplitudes and phases of the individual elements of an antenna array, it is possible to shape the antenna pattern and to steer the direction of the beam electronically.
- The pattern of an array of identical elements is equal to the product of the array factor and the antenna pattern of an individual antenna element.

方程的应用(9.124) 指定增量长度 l 的大小：

$$l = \frac{n_0 u_p}{f_0} = \frac{9 \times 0.8 \times 3 \times 10^8}{10^{10}} = 21.6 \text{ cm}.$$

总之，当 N = 6 和 kd = π 时，方程： (9.121) 变为

$$F_{an}(\gamma') = \frac{\sin^2(3\gamma')}{36 \sin^2(\gamma'/2)}$$

和

$$\gamma' = kd(\cos\theta - \cos\theta_0) = \pi(\cos\theta - \cos\theta_0)$$

和

$$\cos\theta_0 = \frac{2n_0\pi}{kd}\left(\frac{\Delta f}{f_0}\right) = 18\left(\frac{f - 10 \text{ GHz}}{10 \text{ GHz}}\right). \qquad (9.130)$$

阵列方向图的形状与图9-30类似，其主波束方向沿θ = θ_0 。对于 f = f_0 = 10 GHz，θ_0 = 90° (宽边方向)；对于 f = 10.48 GHz，θ_0 = 30°；对于 f = 9.52 GHz，θ_0 = 150°。对于 30° 和 150° 之间的任何其他 θ_0 值，等式： (9.130) 提供了计算振荡器频率 f 所需值的方法。

Concept Question 9-11: Why are antenna arrays useful? Give examples of typical applications.

Concept Question 9-12: Explain how the pattern multiplication principle is used to compute the radiation pattern of an antenna array.

Concept Question 9-13: For a linear array, what roles do the array amplitudes and phases play?

Concept Question 9-14: Explain how electronic beam steering is accomplished.

Concept Question 9-15: Why is frequency scanning an attractive technique for steering the beam of an antenna array?

第9章总结

概念

- 天线是传输线上传播的导波和无界介质中传播的电磁波之间的换能器，反之亦然。
- 除一些由非线性半导体或铁氧体材料构成的固态天线外，天线都是互易器件；它们在传输和接收时表现出相同的辐射方向图。
- 在天线的远场区域，辐射能量近似为平面波。
- 载流天线（例如电线）辐射的电场等于构成天线的所有赫兹偶极子辐射的电场之和。
- 半波偶极子的辐射电阻R_{rad}为73Ω，可以很容易地与传输线匹配。
- 天线的方向特性通过其辐射方向图、方向性、方向图立体角和半功率波束宽度来描述。
- 弗里斯传输公式将天线接收的功率与指定距离处的另一个天线发射的功率联系起来。
- 根据基尔霍夫标量衍射理论，大孔径辐射的远区电场（以波长测量）与孔径上的场分布有关。均匀的孔径分布产生具有尽可能窄的波束宽度的远场方向图。
- 通过控制天线阵列中各个元件的幅度和相位，可以塑造天线方向图并以电子方式控制波束方向。
- 相同元件的阵列的方向图等于阵列因子与单个天线元件的天线方向图的乘积。

Mathematical and Physical Models

Antenna Properties

Pattern solid angle $\Omega_p = \iint_{4\pi} F(\theta,\phi)\, d\Omega$

Directivity $D = \dfrac{4\pi}{\Omega_p}$

Gain $G = \xi D, \quad \xi = \dfrac{P_{rad}}{P_{rad} + P_{loss}}$

Effective area $A_e = \dfrac{\lambda^2 D}{4\pi}$

Far-field distance $R > \dfrac{2d^2}{\lambda}$

Short Dipole ($l \ll \lambda$)

$\widetilde{E}_\theta = \dfrac{jI_0 lk\eta_0}{4\pi}\left(\dfrac{e^{-jkR}}{R}\right)\sin\theta$

$\widetilde{H}_\phi = \dfrac{\widetilde{E}_\theta}{\eta_0}$

$S(R,\theta) = \left(\dfrac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\right)\sin^2\theta$

$D = 1.5$

$\beta = 90^\circ$

$R_{rad} = 80\pi^2 (l/\lambda)^2$

$\lambda/2$ Dipole

$\widetilde{E}_\theta = j60 I_0 \left\{\dfrac{\cos[(\pi/2)\cos\theta]}{\sin\theta}\right\}\left(\dfrac{e^{-jkR}}{R}\right)$

$\widetilde{H}_\phi = \dfrac{\widetilde{E}_\theta}{\eta_0}$

$S(R,\theta) = \dfrac{15 I_0^2}{\pi R^2}\left\{\dfrac{\cos^2[(\pi/2)\cos\theta]}{\sin^2\theta}\right\}$

$D = 1.64$

$\beta = 78^\circ$

$R_{rad} \approx 73\ \Omega$

Friis Transmission Formula

$$\frac{P_{rec}}{P_t} = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 F_t(\theta_t,\phi_t)\, F_r(\theta_r,\phi_r)$$

Rectangular Aperture (Uniform)

$S(R,\theta) = S_0 \operatorname{sinc}^2(\pi l_x \sin\theta/\lambda), \quad x\text{-}z$ plane

$S(R,\theta) = S_0 \operatorname{sinc}^2(\pi l_y \sin\theta/\lambda), \quad y\text{-}z$ plane

$\beta_{xz} = 0.88\dfrac{\lambda}{l_x}, \quad \beta_{yz} = 0.88\dfrac{\lambda}{l_y}$

$D = \dfrac{4\pi A_e}{\lambda^2} \approx \dfrac{4\pi A_p}{\lambda^2}$

Antenna Arrays

Multiplication Principle

$S(R_0,\theta,\phi) = S_e(R_0,\theta,\phi)\, F_a(\theta)$

Uniform Phase

$$F_a(\gamma) = \left|\sum_{i=0}^{N-1} a_i e^{ji\gamma}\right|^2, \text{ with } \gamma = kd\cos\theta = \frac{2\pi d}{\lambda}\cos\theta$$

Linear Phase

$$F_a(\theta) = \left|\sum_{i=0}^{N-1} a_i e^{ji\gamma'}\right|^2, \text{ with } \gamma' = kd\cos\theta - \delta$$

数学与物理模型

天线参数

波束立体角 $\Omega_p = \iint_{4\pi} F(\theta,\phi)\, d\Omega$

方向性系数 $D = \dfrac{4\pi}{\Omega_p}$

增益 $G = \xi D, \quad \xi = \dfrac{P_{rad}}{P_{rad} + P_{loss}}$

有效面积 $A_e = \dfrac{\lambda^2 D}{4\pi}$

远场距离 $R > \dfrac{2d^2}{\lambda}$

短电流元 ($l \ll \lambda$)

$\widetilde{E}_\theta = \dfrac{jI_0 lk\eta_0}{4\pi}\left(\dfrac{e^{-jkR}}{R}\right)\sin\theta$

$\widetilde{H}_\phi = \dfrac{\widetilde{E}_\theta}{\eta_0}$

$S(R,\theta) = \left(\dfrac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\right)\sin^2\theta$

$D = 1.5$

$\beta = 90^\circ$

$R_{rad} = 80\pi^2 (l/\lambda)^2$

半波偶极子

$\widetilde{E}_\theta = j60 I_0 \left\{\dfrac{\cos[(\pi/2)\cos\theta]}{\sin\theta}\right\}\left(\dfrac{e^{-jkR}}{R}\right)$

$\widetilde{H}_\phi = \dfrac{\widetilde{E}_\theta}{\eta_0}$

$S(R,\theta) = \dfrac{15 I_0^2}{\pi R^2}\left\{\dfrac{\cos^2[(\pi/2)\cos\theta]}{\sin^2\theta}\right\}$

$D = 1.64$

$\beta = 78^\circ$

$R_{rad} \approx 73\ \Omega$

弗里斯传输公式

$$\frac{P_{rec}}{P_t} = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 F_t(\theta_t,\phi_t)\, F_r(\theta_r,\phi_r)$$

均匀照射的矩形口径

$S(R,\theta) = S_0 \operatorname{sinc}^2(\pi l_x \sin\theta/\lambda), \quad$ x-z 平面

$S(R,\theta) = S_0 \operatorname{sinc}^2(\pi l_y \sin\theta/\lambda), \quad$ y-z 平面

$\beta_{xz} = 0.88\dfrac{\lambda}{l_x}, \quad \beta_{yz} = 0.88\dfrac{\lambda}{l_y}$

$D = \dfrac{4\pi A_e}{\lambda^2} \approx \dfrac{4\pi A_p}{\lambda^2}$

天线阵列

方向图相乘原理

$S(R_0,\theta,\phi) = S_e(R_0,\theta,\phi)\, F_a(\theta)$

同相激励

$$F_a(\gamma) = \left|\sum_{i=0}^{N-1} a_i e^{ji\gamma}\right|^2, \text{其中 } \gamma = kd\cos\theta = \frac{2\pi d}{\lambda}\cos\theta$$

线性相位递进

$$F_a(\theta) = \left|\sum_{i=0}^{N-1} a_i e^{ji\gamma'}\right|^2, \text{其中 } \gamma' = kd\cos\theta - \delta$$

使用条件：短电流元一栏假设电流沿元长均匀。有效面积式 $A_e=\lambda^2D/(4\pi)$ 对应理想无损天线；一般天线应以增益 G 替代 D，并满足极化及阻抗匹配条件。

Important Terms Provide definitions or explain the meaning of the following terms:

3 dB beamwidth
antenna
antenna array
antenna directivity D
antenna gain G
antenna input impedance
antenna pattern
antenna polarization
aperture distribution
array distribution
array factor $F_a(\theta,\phi)$
azimuth angle
beamwidth β
broadside direction
effective area
(effective aperture) A_e
electronic steering
elevation and azimuth planes
elevation angle
end-fire direction
far-field (or far-zone) region
feeding coefficient
frequency scanning
Friis transmission formula
half-power beamwidth
isotropic antenna
linear phase distribution
loss resistance R_{loss}
null beamwidth
pattern multiplication principle
pattern solid angle Ω_p
power density $\mathbf{S}(R,\theta,\phi)$
Poynting vector
principal planes
radiation efficiency ξ
radiation intensity
(normalized) $F(\theta,\phi)$
radiation lobes
radiation pattern
radiation resistance R_{rad}
reciprocal
scan angle
short dipole (Hertzian dipole)
signal-to-noise ratio S_n
solid angle
spherical propagation factor
steradian
system noise temperature T_{sys}
tapered aperture distribution
zenith angle

PROBLEMS

Sections 9-1 and 9-2: Short Dipole and Antenna Radiation Characteristics

*9.1 A center-fed Hertzian dipole is excited by a current $I_0 = 20$ A. If the dipole is $\lambda/50$ in length, determine the maximum radiated power density at a distance of 1 km.

9.2 A 50-cm long center-fed dipole directed along the z direction and located at the origin is excited by a 1 MHz source. If the current amplitude is $I_0 = 10$ A, determine:

(a) the power density radiated at 2 km along the broadside of the antenna pattern, and

(b) the fraction of the total power radiated within the sector between $\theta = 85°$ and $\theta = 95°$.

9.3 A 1-m long dipole is excited by a 1 MHz current with an amplitude of 12 A. What is the average power density radiated by the dipole at a distance of 5 km in a direction that is 45° from the dipole axis?

*9.4 Determine:

(a) the direction of maximum radiation,

(b) directivity,

(c) beam solid angle, and

(d) half-power beamwidth in the x–z plane,

for an antenna whose normalized radiation intensity is given by

$$F(\theta,\phi) = \begin{cases} 1 & \text{for } 0 \le \theta \le 60° \text{and } 0 \le \phi \le 2\pi \\ 0 & \text{elsewhere.} \end{cases}$$

Suggestion: Sketch the pattern prior to calculating the desired quantities.

9.5 Repeat Problem 9.4 for an antenna with

$$F(\theta,\phi) = \begin{cases} \sin^2\theta\cos^2\phi & \text{for } 0 \le \theta \le \pi \\ & \text{and } -\pi/2 \le \phi \le \pi/2 \\ 0 & \text{elsewhere.} \end{cases}$$

9.6 A 2-m long center-fed dipole antenna operates in the AM broadcast band at 1 MHz. The dipole is made of copper wire with a radius of 1 mm.

(a) Determine the radiation efficiency of the antenna.

*(b) What is the antenna gain in decibels?

(c) What antenna current is required so that the antenna will radiate 80 W, and how much power will the generator have to supply to the antenna?

9.7 Repeat Problem 9.6 for a 20-cm long antenna operating at 5 MHz.

9.8 Determine the frequency dependence of the radiation efficiency of the short dipole, and plot it over the range from 600 kHz to 60 MHz. The dipole is made of copper, its length is 10 cm, and its circular cross section has a radius of 1 mm.

*Answer(s) available in Appendix E.

428 CHAPTER 9 RADIATION AND ANTENNAS

Important Terms Provide definitions or explain the meaning of the following terms:

3 dB beamwidth
antenna
antenna array
antenna directivity D
antenna gain G
antenna input impedance
antenna pattern
antenna polarization
aperture distribution
array distribution
array factor $F_a(\theta,\phi)$
azimuth angle
beamwidth β
broadside direction
effective area
(effective aperture) A_e
electronic steering
elevation and azimuth planes
elevation angle
end-fire direction
far-field (or far-zone) region
feeding coefficient
frequency scanning
Friis transmission formula
half-power beamwidth
isotropic antenna
linear phase distribution
loss resistance R_{loss}
null beamwidth
pattern multiplication principle
pattern solid angle Ω_p
power density $\mathbf{S}(R,\theta,\phi)$
Poynting vector
principal planes
radiation efficiency ξ
radiation intensity
(normalized) $F(\theta,\phi)$
radiation lobes
radiation pattern
radiation resistance R_{rad}
reciprocal
scan angle
short dipole (Hertzian dipole)
signal-to-noise ratio S_n
solid angle
spherical propagation factor
steradian
system noise temperature T_{sys}
tapered aperture distribution
zenith angle

问题

第 9-1 和 9-2 节：短偶极子和天线辐射特性

*9.1 中心馈电赫兹偶极子由电流 $I_0 = 20$ A 激励。如果偶极子长度为 $\lambda/50$，则确定 1 km 距离处的最大辐射功率密度。

9.2 沿 z 方向定向并位于原点的 50 厘米长的中心馈电偶极子由 1 MHz 源激励。如果电流幅度为 $I_0 = 10$ A，则确定：

(a) 沿天线方向图宽边 2 km 处辐射的功率密度，以及 (b) $\theta = 85°$ 和 $\theta = 95°$ 之间扇区内辐射总功率的比例。

9.3 一个 1 米长的偶极子被 1 MHz 电流激励，幅度为 12 A。该偶极子在距离偶极子轴 45° 方向 5 公里处辐射的平均功率密度是多少？

*9.4 确定：(a) 最大辐射方向，(b) 方向性，(c) 波束立体角，(d) x-z 平面内的半功率波束宽度，

对于其归一化辐射强度由下式给出的天线

$$F(\theta,\phi) = \begin{cases} 1 & \text{for } 0 \le \theta \le 60° \text{and } 0 \le \phi \le 2\pi \\ 0 & \text{elsewhere.} \end{cases}$$

建议：在计算所需数量之前绘制图案草图。

9.5 对天线重复问题 9.4

$$F(\theta,\phi) = \begin{cases} \sin^2\theta\cos^2\phi & \text{for } 0 \le \theta \le \pi \\ & \text{and } -\pi/2 \le \phi \le \pi/2 \\ 0 & \text{elsewhere.} \end{cases}$$

9.6 2 米长的中心馈电偶极天线在 1 MHz 的 AM 广播频段工作。偶极子由半径为1毫米的铜线制成。

(a) 确定天线的辐射效率。

*(b) 天线增益（以分贝为单位）是多少？ (c) 需要多大的天线电流才能使天线辐射 80 W，发电机必须向天线提供多少功率？

9.7 对于工作频率为 5 MHz 的 20 厘米长天线，重复问题 9.6。

9.8 确定短偶极子辐射效率的频率依赖性，并在 600 kHz 至 60 MHz 范围内绘制它。偶极子由铜制成，长度为10厘米，圆形横截面的半径为1毫米。

*Answer(s) available in Appendix E.

*9.9 An antenna with a pattern solid angle of 1.5 (sr) radiates 60 W of power. At a range of 1 km, what is the maximum power density radiated by the antenna?

9.10 An antenna with a radiation efficiency of 90% has a directivity of 7.0 dB. What is its gain in decibels?

*9.11 The radiation pattern of a circular parabolic-reflector antenna consists of a circular major lobe with a half-power beamwidth of 3° and a few minor lobes. Ignoring the minor lobes, obtain an estimate for the antenna directivity in dB.

9.12 The normalized radiation intensity of a certain antenna is given by

$$F(\theta) = \exp(-20\theta^2) \qquad \text{for } 0 \le \theta \le \pi$$

where θ is in radians. Determine:

(a) the half-power beamwidth,

(b) the pattern solid angle, and

(c) the antenna directivity.

Sections 9-3 and 9-4: Dipole Antennas

9.13 Repeat Problem 9.6 for a 1-m long half-wave dipole that operates in the FM/TV broadcast band at 150 MHz.

*9.14 Assuming the loss resistance of a half-wave dipole antenna to be negligibly small and ignoring the reactance component of its antenna impedance, calculate the standing-wave ratio on a 50 Ω transmission line connected to the dipole antenna.

9.15 A 50-cm long dipole is excited by a sinusoidally varying current with an amplitude $I_0 = 5$ A. Determine the total radiated power if the oscillating frequency is:

(a) 1 MHz,

(b) 300 MHz.

9.16 For a short dipole with length l where $l \ll \lambda$, instead of treating the current $\tilde{I}(z)$ as constant along the dipole, as was done in Section 9-1, a more realistic approximation that ensures the current goes to zero at the dipole ends is to describe $\tilde{I}(z)$ by the triangular function

$$\tilde{I}(z) = \begin{cases} I_0(1 - 2z/l) & \text{for } 0 \le z \le l/2 \\ I_0(1 + 2z/l) & \text{for } -l/2 \le z \le 0 \end{cases}$$

shown in Fig. P9.16. Use this current distribution to determine the following:

*(a) the far-field $\tilde{\mathbf{E}}(R, \theta, \phi)$,

(b) the power density $S(R, \theta, \phi)$,

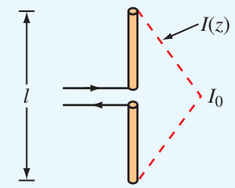


Figure P9.16 Triangular current distribution on a short dipole (Problem 9.16).

(c) the directivity D, and

(d) the radiation resistance R_{rad}.

9.17 For a dipole antenna of length $l = 3\lambda/2$,

*(a) determine the directions of maximum radiation,

(b) obtain an expression for $S_{\max}$,

(c) generate a plot of the normalized radiation pattern $F(\theta)$, and

(d) compare your pattern with that shown in Fig. 9-17(c).

9.18 For a dipole antenna of length $l = \lambda/4$,

(a) determine the directions of maximum radiation,

(b) obtain an expression for $S_{\max}$, and

(c) generate a plot of the normalized radiation pattern $F(\theta)$.

9.19 Repeat parts (a)–(c) of Problem 9.17 for a dipole of length $l = 3\lambda/4$.

*9.20 Repeat parts (a)–(c) of Problem 9.17 for a dipole of length $l = \lambda$.

9.21 A car antenna is a vertical monopole over a conducting surface. Repeat Problem 9.6 for a 1-m long car antenna operating at 1 MHz. The antenna wire is made of aluminum with $\mu_c = \mu_0$ and $\sigma_c = 3.5 \times 10^7$ S/m, and its diameter is 1 cm.

9.22 For a dipole antenna of length $l = 2.5\lambda$, use Module 9.2 to determine:

(a) approximate directions of maximum radiation,

(b) antenna directivity, and

(c) antenna radiation resistance.

9.23 For a dipole antenna of length $l = 3\lambda$, use Module 9.2 to determine:

(a) approximate directions of maximum radiation,

(b) antenna directivity, and

(c) antenna radiation resistance.

*9.9 方向图立体角为 1.5 (sr) 的天线辐射 60 W 功率。在1公里范围内，天线辐射的最大功率密度是多少？

9.10 辐射效率为 90% 的天线的方向性为 7.0 dB。它的增益是多少分贝？

*9.11 圆形抛物面反射器天线的辐射方向图由半功率波束宽度为 3° 的圆形主瓣和几个小瓣组成。忽略小瓣，获得天线方向性的估计值（以 dB 为单位）。

9.12 某个天线的归一化辐射强度由下式给出

$$F(\theta) = \exp(-20\theta^2) \qquad \text{for } 0 \le \theta \le \pi$$

其中 θ 的单位是弧度。决定：

(a) 半功率波束宽度，

(b) 图案立体角，以及

(c) 天线方向性。

第 9-3 节和第 9-4 节：偶极天线

9.13 对于工作在 150 MHz FM/TV 广播频段的 1 米长半波偶极子，重复问题 9.6。

*9.14 假设半波偶极子天线的损耗电阻小到可以忽略不计，并忽略其天线阻抗的电抗分量，计算连接到偶极子天线的 50 Ω 传输线上的驻波比。

9.15 A 50 厘米长的偶极子由振幅为 $I_0 = 5$ A 的正弦变化电流激励。如果振荡频率为：，则确定总辐射功率：

(a) 1 MHz，(
b) 300 MHz
。

9.16 对于长度为 l 的短偶极子，其中 $l \ll \lambda$，不是将电流 $\tilde{}$I(z) 视为沿偶极子的常数，如第 9-1 节中所做的那样，确保电流在偶极子末端变为零的更现实的近似是通过三角函数来描述 $\tilde{}$I(z)

$$\tilde{I}(z) = \begin{cases} I_0(1 - 2z/l) & \text{for } 0 \le z \le l/2 \\ I_0(1 + 2z/l) & \text{for } -l/2 \le z \le 0 \end{cases}$$

如图P9.16所示。使用此电流分布可以确定以下内容：

*(a) 远场 $\tilde{}$E(R, θ , ϕ)，

(b) 功率密度 S(R, θ , ϕ)，

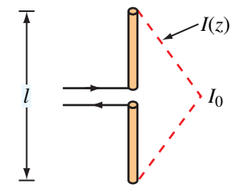


图 P9.16 短偶极子上的三角形电流分布（问题 9.16）。

(c)方向性D，以及

(d)辐射电阻R_{rad}。

9.17 对于长度为 $l = 3\lambda/2$ 的偶极天线，

*(a) 确定最大辐射方向，

(b) 获得 $S_{\max}$ 的表达式，(c) 生成归一化辐射方向图 F(θ) 的图，以及 (d) 将您的方向图与图 9-17(c) 中所示的方向图进行比较。

9.18 对于长度为 $l = \lambda/4$ 的偶极天线，

(a) 确定最大辐射方向，

(b) 获得 $S_{\max}$ 的表达式，并且

(c) 生成归一化辐射方向图 F(θ)。

9.19 对于长度为 $l = 3\lambda/4$ 的偶极子，重复问题 9.17 的 (a)–(c) 部分。

*9.20 对于长度为 $l = \lambda$ 的偶极子，重复问题 9.17 的 (a)–(c) 部分。

9.21 汽车天线是导电表面上的垂直单极天线。对于工作频率为 1 MHz 的 1 米长汽车天线，重复问题 9.6。天线线材质为$\mu_c = \mu_0$和$\sigma_c = 3.5 \times 10^7$ S/m的铝，直径为1 cm。

9.22 对于长度为 $l = 2.5\lambda$ 的偶极天线，使用模块 9.2 确定：

(a) 最大辐射的大致方向，

(b) 天线方向性，以及 (c) 天线辐射电阻。

9.23 对于长度为 $l = 3\lambda$ 的偶极天线，使用模块 9.2 确定：

(a) 最大辐射的大致方向，

(b) 天线方向性，以及

(c) 天线辐射电阻。

9.24 A dipole antenna has a radiation impedance

$$Z_{\text{rad}} = R_{\text{rad}} + jX_{\text{rad}}.$$

For a dipole of length $l = \lambda/2$, $R_{\text{rad}} = 73.13\ \Omega$ and $X_{\text{rad}} = 42.55\ \Omega$. Use Module 9.3 to modify the value of l slightly so as to reduce X_{rad} to less than 1 Ω.

(a) What is the value of l?

(b) What is the corresponding value of R_{rad}?

Sections 9-5 and 9-6: Effective Area and Friis Formula

9.25 Determine the effective area of a half-wave dipole antenna at 100 MHz, and compare it with its physical cross section if the wire diameter is 2 cm.

*__9.26__ A 3 GHz line-of-sight microwave communication link consists of two lossless parabolic dish antennas, each 1 m in diameter. If the receive antenna requires 10 nW of receive power for good reception and the distance between the antennas is 40 km, how much power should be transmitted?

9.27 A half-wave dipole TV broadcast antenna transmits 1 kW at 50 MHz. What is the power received by a home television antenna with a 3 dB gain if located at a distance of 30 km?

*__9.28__ A 150 MHz communication link consists of two vertical half-wave dipole antennas separated by 2 km. The antennas are lossless, the signal occupies a bandwidth of 3 MHz, the system noise temperature of the receiver is 600 K, and the desired signal-to-noise ratio is 17 dB. What transmitter power is required?

9.29 Consider the communication system shown in **Fig. P9.29** with all components properly matched. Determine the following if $P_t = 10$ W and $f = 6$ GHz.

(a) What is the power density at the receiving antenna (assuming proper alignment of antennas)?

(b) What is the received power?

(c) If $T_{\text{sys}} = 1{,}000$ K and the receiver bandwidth is 20 MHz, what is the signal-to-noise ratio in decibels?

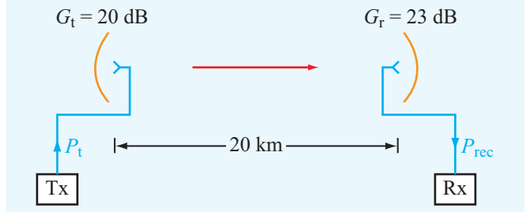


Figure P9.29 Communication system of Problem 9.29.

9.30 The configuration shown in **Fig. P9.30** depicts two vertically oriented half-wave dipole antennas pointed towards each other with both positioned on 100-m tall towers separated by a distance of 5 km. If the transit antenna is driven by a 50 MHz current with amplitude $I_0 = 2$ A, determine the following:

*__(a)__ The power received by the receive antenna in the absence of the surface. (Assume both antennas to be lossless.)

(b) The power received by the receive antenna after incorporating reflection by the ground surface, assuming the surface to be flat and to have $\varepsilon_r = 9$ and conductivity $\sigma = 10^{-3}$ (S/m).

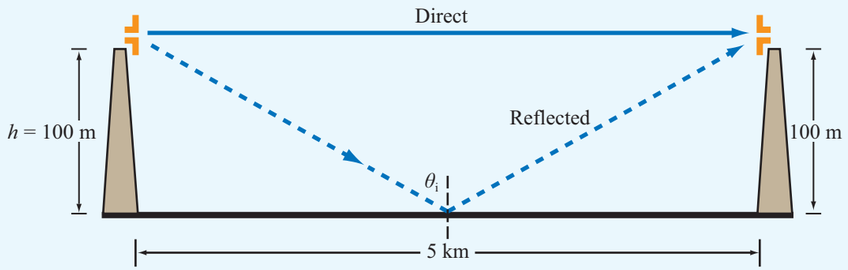


Figure P9.30 Problem 9.30.

430 CHAPTER 9 RADIATION AND ANTENNAS

9.24 偶极天线具有辐射阻抗

$$Z_{\text{rad}} = R_{\text{rad}} + jX_{\text{rad}}.$$

对于长度为 $l = \lambda/2$ 的偶极子，$R_{\text{rad}} = 73.13\ \Omega$ 和 $X_{\text{rad}} = 42.55\ \Omega$。使用模块9.3稍微修改l的值，使$X_{\text{rad}}$减少到小于1Ω。

(a) l 的值是多少？ (b) R_{rad} 对应的值是多少？

第 9-5 节和第 9-6 节：有效面积和弗里斯公式

9.25 确定半波偶极子天线在 100 MHz 下的有效面积，并与线径为 2 cm 时的物理横截面进行比较。

*9.26 3 GHz 视距微波通信链路由两个无损抛物面碟形天线组成，每个直径为 1 m。如果接收天线需要 10 nW 的接收功率才能获得良好的接收效果，并且天线之间的距离为 40 km，则应发射多少功率？

9.27 半波偶极子电视广播天线在 50 MHz 下发射 1 kW 功率。如果位于 30 公里的距离，增益为 3 dB 的家用电视天线接收到的功率是多少？

*9.28 150 MHz 通信链路由两个相距 2 km 的垂直半波偶极天线组成。天线无损，信号占用带宽3 MHz，接收机系统噪声温度600 K，期望信噪比17 dB。需要多大的发射机功率？

9.29 考虑图 P9.29 所示的通信系统，所有组件均正确匹配。如果 $P_t = 10$ W 和 f = 6 GHz，请确定以下内容。

(a) 接收天线的功率密度是多少（假设天线正确对准）？ (b) 接收功率是多少？ (c) 如果 $T_{\text{sys}} = 1{,}000$ K 且接收器带宽为 20 MHz，则信噪比（以分贝为单位）是多少？

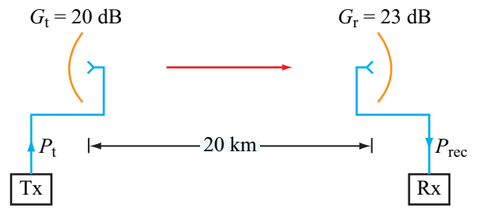


图 P9.29 问题 9.29 的通信系统。

9.30 图 P9.30 所示的配置描绘了两个垂直定向的半波偶极子天线，它们彼此指向，且均位于 100 米高的塔上，塔上相距 5 公里。如果中转天线由幅度为 $I_0 = 2$ A 的 50 MHz 电流驱动，请确定以下内容：

*(a) 在没有表面的情况下接收天线接收到的功率。 （假设两个天线都是无损的。） (b) 接收天线在纳入地面反射后接收到的功率，假设地面平坦且具有 $\varepsilon_r = 9$ 和电导率 $\sigma = 10^{-3}$ (S/m)。

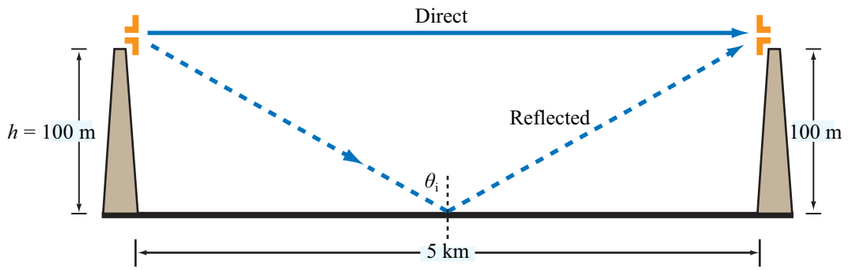


图 P9.30 问题 9.30。